In [4]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

np.random.seed(42)

dates = pd.date_range(start='2025-01-01', periods=180, freq='D')
x = np.arange(180)

data = pd.DataFrame({
    'Date': dates,
    'Page_Views': 200 + x * 2 + 50 * np.sin(2 * np.pi * x / 30) + np.random.norm
    'Unique_Visitors': 100 + x + 20 * np.sin(2 * np.pi * x / 30) + np.random.nor
    'Bounce_Rate': 40 + 5 * np.sin(2 * np.pi * x / 7) + np.random.normal(0, 2, 1
})

data.set_index('Date', inplace=True)
data = data.interpolate(method='linear')

print(data.head())

decomposition = seasonal_decompose(data['Page_Views'], model='additive', period=
decomposition.plot()
plt.show()

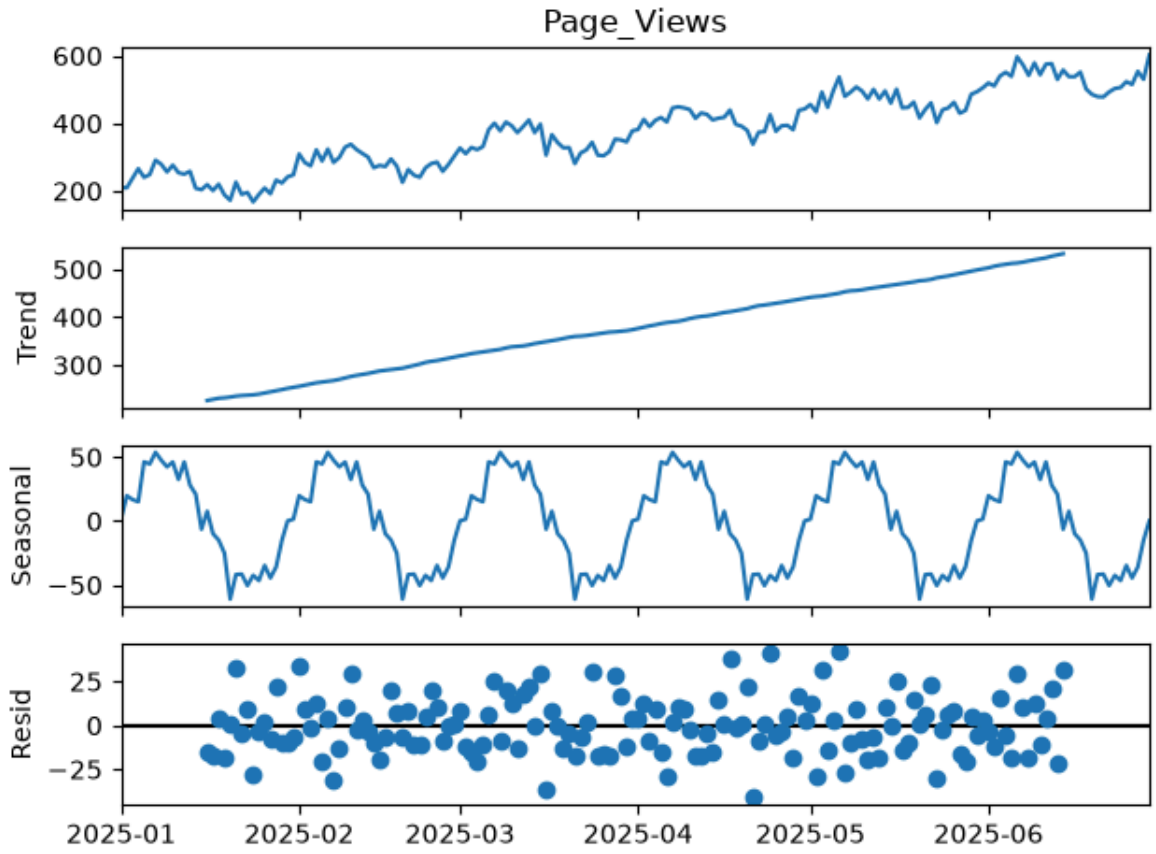
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
data[['Page_Views', 'Page_Views_MA']].plot(figsize=(12, 5))
plt.title('Page Views with Moving Average')
plt.show()

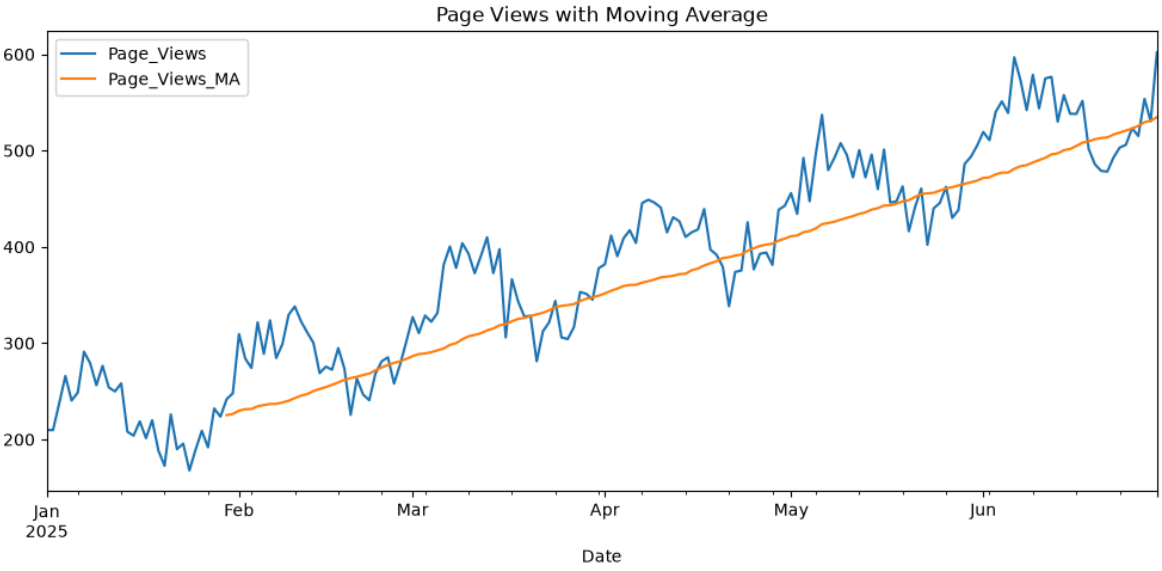
data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) / data['
anomalies = data[data['Page_Views_Z'].abs() > 3]

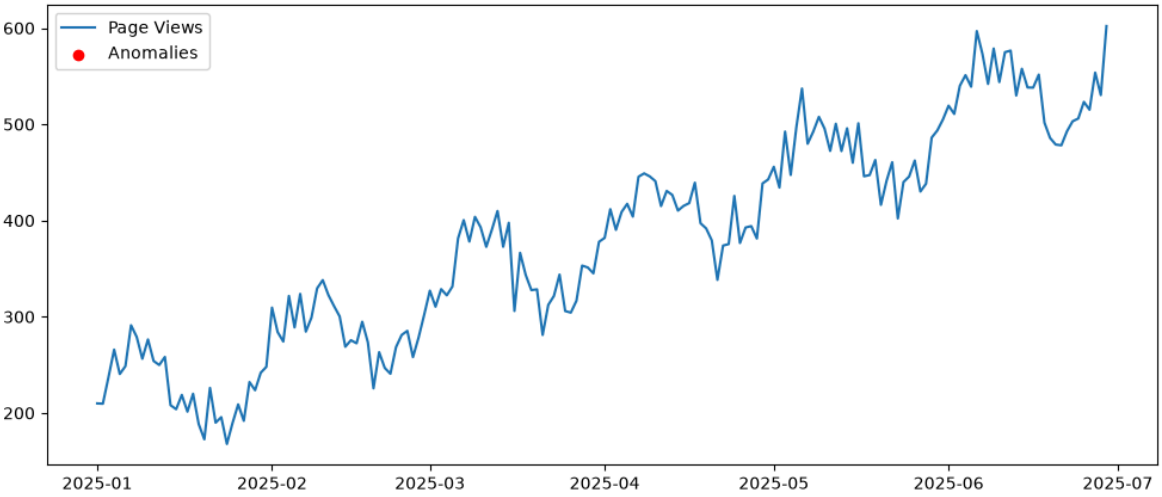
plt.figure(figsize=(12, 5))
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomal
plt.legend()
plt.show()

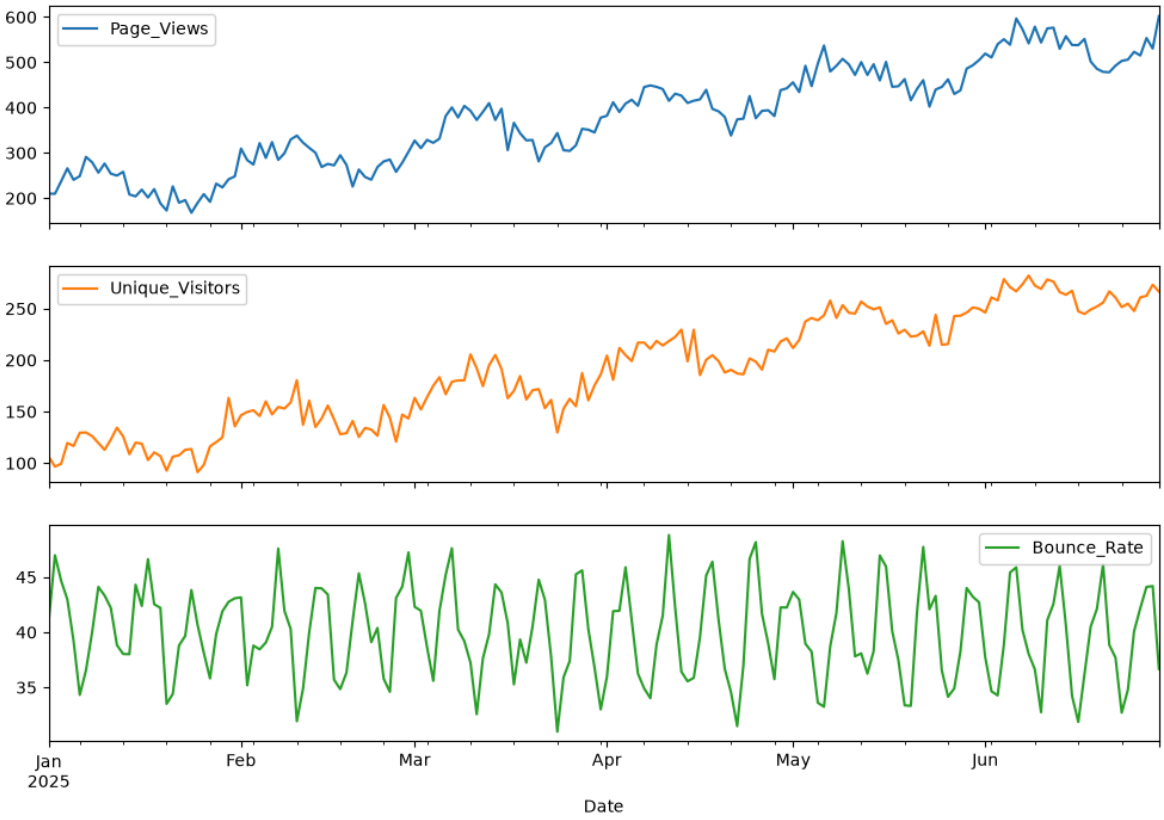
data[['Page_Views', 'Unique_Visitors', 'Bounce_Rate']].plot(subplots=True, figsi
plt.show()
```

```
            Page_Views  Unique_Visitors  Bounce_Rate
Date
2025-01-01  209.934283       106.256673    41.038693
2025-01-02  209.630299        96.586658    46.974635
2025-01-03  237.290603        99.425808    44.657119
2025-01-04  265.849860       119.580429    42.972842
2025-01-05  240.474174       116.628269    39.210869
```


Page_Views
600
400
200
Trend
500
400
300
Seasonal
50
0
−50
Resid
25
0
−25
2025-01
2025-02
2025-03
2025-04
2025-05
2025-06


Page Views with Moving Average
Page_Views
Page_Views_MA
600
500
400
300
200
Jan
2025
Feb
Mar
Apr
May
Jun
Date


Page Views
Anomalies
600
500
400
300
200
2025-01
2025-02
2025-03
2025-04
2025-05
2025-06
2025-07


Page_Views
600
500
400
300
200
Unique_Visitors
250
200
150
100
Bounce_Rate
45
40
35
Jan
2025
Feb
Mar
Apr
May
Jun
Date

In [ ]: